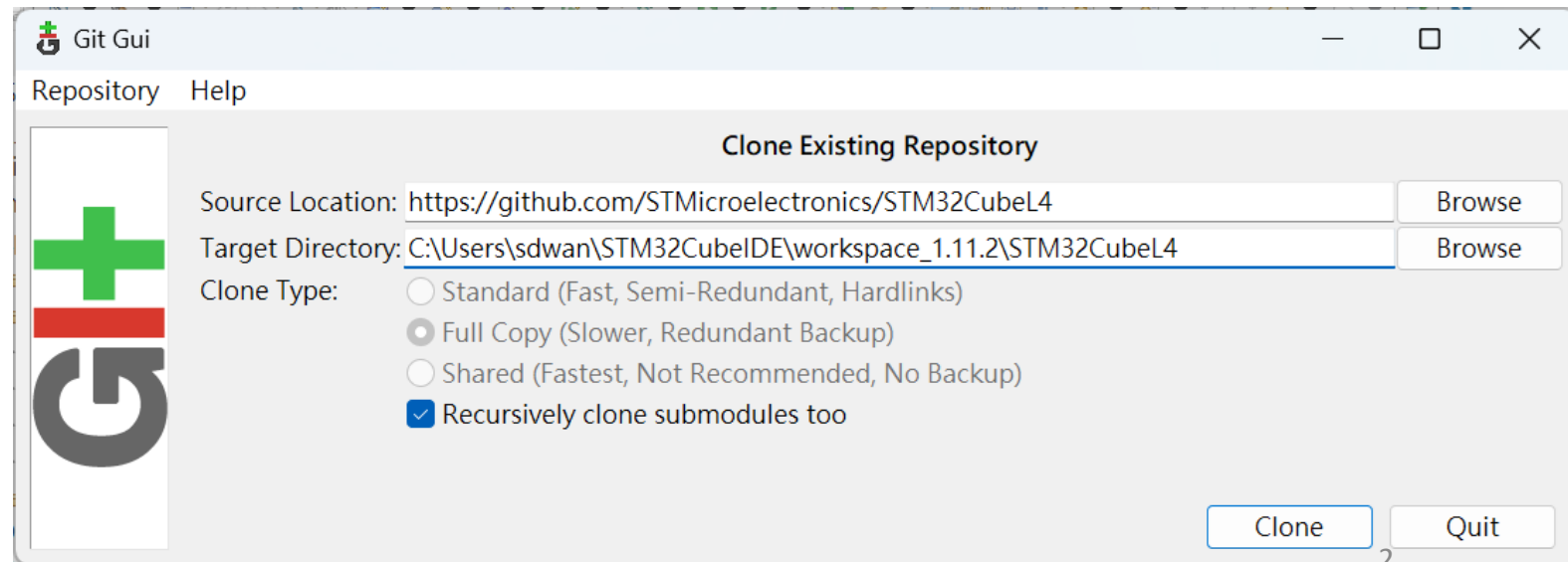


STM32CubeIDE: wifi example

Lab: wifi client/server example

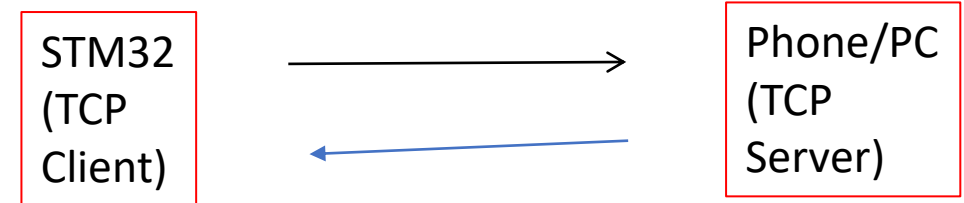
WIFI_CLIENT_SERVER example

- STM32CubeL4 firmware
 - <https://github.com/STMicroelectronics/STM32CubeL4>
 - git clone <https://github.com/STMicroelectronics/STM32CubeL4>
 - Or use git gui
- Check source code:
 - <https://github.com/STMicroelectronics/STM32CubeL4/Projects/B-L475E-IOT01A/Applications/WiFi>
- Two wifi examples
 - WiFi_Client_Server
 - WiFi_HTTP_Server



Application Description from readme.txt

- This application shows how to use the **es-wifi module** to perform a TCP client mode using STM32Cube HAL. It demonstrates how to set up a client program and connect it to a TCP server.
- The Inventek es-WiFi module is offered with a complete TCP/IP stack and requires only a simple AT command over the **SPI** interface to communicate with the module.
- In a typical embedded application, the STM32 microcontroller is used to send the AT command to initiate a client connection.
- The es-wifi module starts to connect to an **access point**. Once connected, we should connect to a **specific server**. The address of the server is declared in **RemoteIP[] in main.c**.
- Then, the es-wifi module will be initiated as a client and connected to the server (provided by a smartphone) and will check the connection. 8002 is the port number used for access.
- This application may be used with a **Smartphone** (more detailed setup instructions are described in section "Hardware and Software environment").



In main.c define
Server IP and Port

Import/open project from file systems

IDE

Import Projects from File System or Archive

Import Projects from File System or Archive

This wizard analyzes the content of your folder or archive file to find projects and import them in the IDE.

Import source: C:\Users\sdwan\STM32CubeIDE\workspace_1.11.2\STM32CubeL4\Projects\B-L475E-IOT01A\Applications\WiFi\WiFi_Client_Server\STM32CubeIDE

Directory...

Archive...

type filter text

Folder

Import as

☒ STM32CubeIDE

STM32CubeIDE...

Select All

Deselect All

1 of 1 selected

☐ Hide already open projects

☐ Close newly imported projects upon completion

Use [installed project configurators](#) to:

☒ Search for nested projects

☒ Detect and configure project natures

Working sets

☐ Add project to working sets

Working sets: IOTA1_freertos

New...

Select...

[Show other specialized import wizards](#)

?

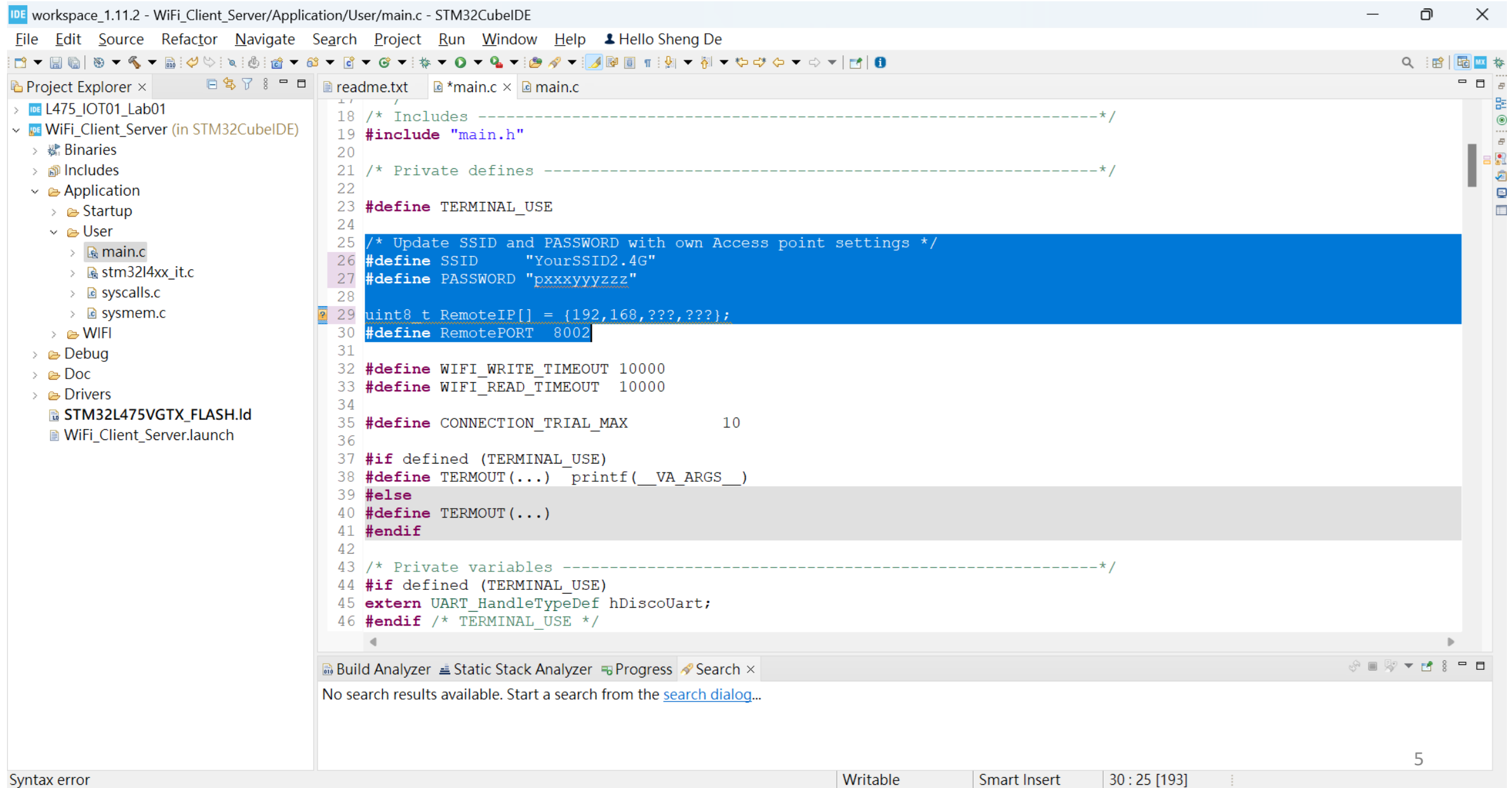
< Back

Next >

Finish

Cancel

Check main.c and edit some #define



The screenshot shows the STM32CubeIDE interface. The Project Explorer on the left shows the project structure: L475_IOT01_Lab01 > WiFi_Client_Server (in STM32CubeIDE) > Application > User > main.c. The main.c file is open in the editor, showing the following code:

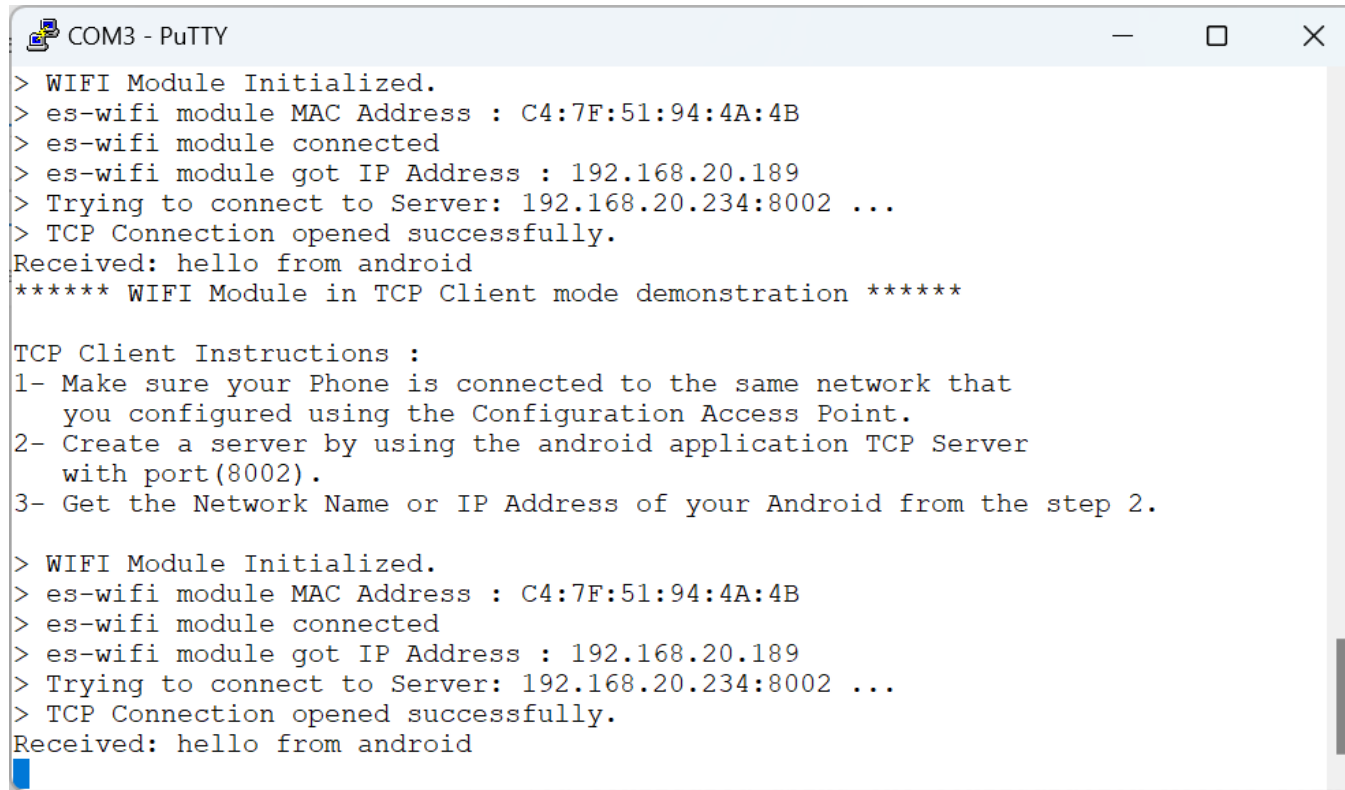
```
17 /* Includes -----*/
18 #include "main.h"
19
20 /* Private defines -----*/
21
22 #define TERMINAL_USE
23
24 /* Update SSID and PASSWORD with own Access point settings */
25 #define SSID "YourSSID2.4G"
26 #define PASSWORD "pxxxxxyyyzzz"
27
28 uint8_t RemoteIP[] = {192,168,???,???};
29 #define RemotePORT 8002
30
31
32 #define WIFI_WRITE_TIMEOUT 10000
33 #define WIFI_READ_TIMEOUT 10000
34
35 #define CONNECTION_TRIAL_MAX 10
36
37 #if defined (TERMINAL_USE)
38 #define TERMOUT(...) printf(__VA_ARGS__)
39 #else
40 #define TERMOUT(...)
41 #endif
42
43 /* Private variables -----*/
44 #if defined (TERMINAL_USE)
45 extern UART_HandleTypeDef hDiscoUart;
46 #endif /* TERMINAL_USE */
```

The following lines are highlighted in blue in the original image:

- Line 25: `/* Update SSID and PASSWORD with own Access point settings */`
- Line 26: `#define SSID "YourSSID2.4G"`
- Line 27: `#define PASSWORD "pxxxxxyyyzzz"`
- Line 29: `uint8_t RemoteIP[] = {192,168,???,???};`
- Line 30: `#define RemotePORT 8002`

The bottom status bar shows "Syntax error" on the left and "Writable", "Smart Insert", and "30 : 25 [193]" on the right. The bottom right corner of the window displays the number "5".

Run the project – connection opened successfully - using Putty as serial console

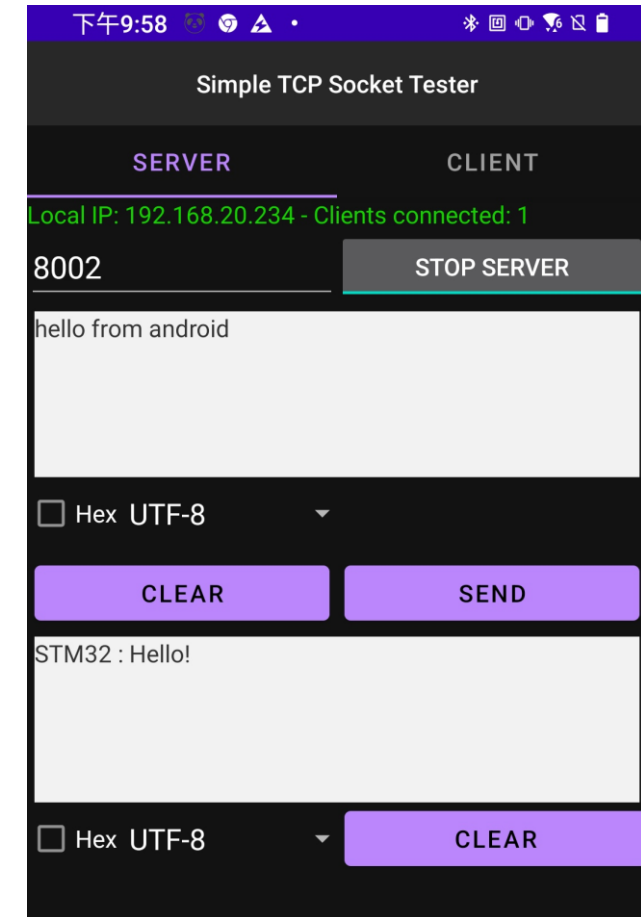


```
COM3 - PuTTY
> WIFI Module Initialized.
> es-wifi module MAC Address : C4:7F:51:94:4A:4B
> es-wifi module connected
> es-wifi module got IP Address : 192.168.20.189
> Trying to connect to Server: 192.168.20.234:8002 ...
> TCP Connection opened successfully.
Received: hello from android
***** WIFI Module in TCP Client mode demonstration *****

TCP Client Instructions :
1- Make sure your Phone is connected to the same network that
   you configured using the Configuration Access Point.
2- Create a server by using the android application TCP Server
   with port(8002).
3- Get the Network Name or IP Address of your Android from the step 2.

> WIFI Module Initialized.
> es-wifi module MAC Address : C4:7F:51:94:4A:4B
> es-wifi module connected
> es-wifi module got IP Address : 192.168.20.189
> Trying to connect to Server: 192.168.20.234:8002 ...
> TCP Connection opened successfully.
Received: hello from android
```

STM32 (tcp client)



Android phone (tcp server)